

Identification of functional group

Experiment No. 18

Date: / /

Aim: To determine the functional group of the given organic compound.

Theory: Organic chemistry is the study of hydrocarbons and its derivatives. Organic compounds are classified on the basis of their functional groups.

Functional Group: An atom or group of atoms bonded together in a unique manner so as to form a compound and represent its properties is called functional group.

The physical and chemical properties of compound containing different functional groups are different. So identification of functional group for an unknown organic compound is based on physical and chemical properties.

Apparatus: Test tube, test tube holder, test tube stand, etc.

Note : Only one compound should be given for identification of only one functional group in practical examination.

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO ₃ solution.	Substance soluble with brisk effervescence of CO ₂ gas	Carboxylic group present $\left(\begin{array}{c} \text{O} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{OH} \end{array} \right)$
	No brisk effervescence of CO ₂ gas	Carboxylic group absent
2. Test for Phenolic group Substance + little water + neutral FeCl ₃ solution	Violet/green/blue/red (any one) colouration	Phenolic (Ar-OH) group present
	No Violet/green/blue/red (any one) colouration	Phenolic group absent
3. Test for Aldehyde group (liquid) Substance + 2-3 mL Schiff's reagent shake well	Pink colour appears	Aldehyde $\left(\begin{array}{c} \text{O} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{H} \end{array} \right)$ group present
	No pink colouration	Aldehyde group absent
OR		
Substance + Tollen's reagent heat/warm	Silver mirror is formed on inner side of the test tube	Aldehyde group $\left(\begin{array}{c} \text{O} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{H} \end{array} \right)$ present
	Silver mirror is not formed	Aldehyde group absent
4. Test for Ketonic group (liquid) Substance + few drops of NaOH + few drops of sodium nitroprusside solution Na ₂ [Fe(CN) ₅ NO]	Red colour	Ketonic group $\left(>\text{C}=\text{O} \right)$ group present.
	No red colour	Ketonic group absent.

5. Test for Amino group

Substance + conc. HCl in test tube -I and shake to dissolve. Cool under tap water, add excess of NaNO_2 .

In another test tube - II, take β -naphthol in NaOH

Add cold solution from test tube -I into β -naphthol in NaOH from test tube -II.

Orange red dye stuff is formed

Amino $\left(\begin{array}{c} \text{H} \\ \diagup \text{N} \diagdown \\ \text{H} \end{array} \right)$ group present.

No orange red dye stuff

Amino group absent

6. Test for Unsaturation/ Hydrocarbons (solid)

If all the above tests (from 1 to 5) are negative and the substance is water immiscible or insoluble

No specific test or functional group then

Unsaturated hydrocarbon is present [Unsaturation $\left(\text{>C=C<} \right)$ or $\left(-\text{C}\equiv\text{C}- \right)$ present]

OR

Alkaline KMnO_4 Test (Bayer's Test)

Organic substance + little water + dil. alkaline KMnO_4 solution (1%)

Decolourisation (pink colour disappears)

Unsaturation present

No decolourisation

Saturation present

[Alkaline KMnO_4 (1%): Dissolve 1 g of KMnO_4 (solid) in 100 ml of distilled water. Now add 10 g of anhydrous Na_2CO_3 . Shake to dissolve and stopper the bottle]

7. Test for alcoholic group (liquid)

Substance + 2-3 crystals of $\text{K}_2\text{Cr}_2\text{O}_7$ + dil. H_2SO_4 . Boil and hold AgNO_3 paper near mouth of test tube (Alcohols having sweet smell and they are miscible in water)

Black stains on AgNO_3 paper

Alcoholic (R-OH) group present

No black stains on AgNO_3 paper

Alcoholic group absent

OR

Substance + a very small piece of sodium metal

effervescence of H_2 gas

Alcoholic (R-OH) group present

No effervescence of H_2 gas

Alcoholic group absent

Result: The given organic compound contains following functional group.

Name	Structure of functional group
.....	

Compound No.1

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO ₃ solution.	Substance soluble with CO ₂ gas.	carboxylic group is present.

Result: The given organic compound no.1 contains following functional group.

Name of the functional group	Structure of functional group
carboxylic acid group	$-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{OH}$

Compound No.2

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO ₃ solution.	No brisk effervescence of CO ₂ .	carboxylic group absent.
2. Test for Phenolic group Substance + little water + neutral FeCl ₃ solution	violet colouration.	phenolic (Ar-OH) group present.

Result: The given organic compound no.2 contains following functional group.

Name of the functional group	Structure of functional group
phenolic (Ar-OH)	Ar-OH

Compound No.3

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO ₃ solution.	No effervescence of CO ₂ .	carboxylic group absent.
2. Test for Phenolic group Substance + little water + neutral FeCl ₃ solution	No colouration.	phenolic group absent.
3. Test for Aldehyde group (liquid) Substance + 2-3 mL Schiff's reagent/ Tollen's reagent and heat	pink colour appears.	Aldehyde group present.

Result: The given organic compound no.3 contains following functional group.

Name of the functional group	Structure of functional group
Aldehyde group	$-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$

Compound No.4

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO ₃ solution.	No effervescence of CO ₂ .	carboxylic group absent.

Test	Observation	Inference
2. Test for Phenolic group Substance + little water + neutral FeCl_3 solution	No colouration	phenolic group absent.
3. Test for Aldehyde group (liquid) Substance + 2-3 mL Schiff's reagent/ Tollen's reagent and heat	No colouration	Aldehyde group absent.
4. Test for Ketone group (liquid) Substance + few drops of NaOH + few drops of sodium nitroprusside soln	Red colouration	Ketonic group is present.

Result: The given organic compound no.4 contains following functional group.

Name of the functional group	Structure of functional group
.....Ketonic.....	$\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—}$

Compound No.5

Test	Observation	Inference
1. Test for Carboxylic acid group Substance + 10% NaHCO_3 solution.	No effervances of CO_2	—COOH group is absent.
2. Test for Phenolic group Substance + little water + neutral FeCl_3 soln	No colouration	Ph-OH absent
3. Test for Aldehyde group (liquid) Substance + 2-3 mL Schiff's reagent/ Tollen's reagent and heat	No colouration	$\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—H}$ absent
4. Test for Ketonic group (liquid) Substance + few drops of NaOH + few drops of sodium nitroprusside soln	No colouration	$\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—}$ absent
5. Test for Amine group Substance + conc. HCl and shake to dissolve. Cool under tap water, add excess of NaNO_2 + β -naphthol in NaOH	orange red dye stuff form	Amine (—NH_2) group present

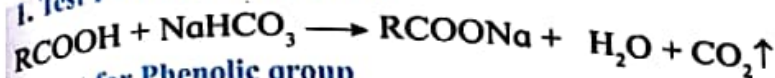
Result: The given organic compound no.5 contains following functional group.

Name of the functional group	Structure of functional group
Amine	—NH_2

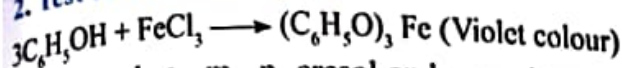
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ons involved :

1. Test For Carboxylic group

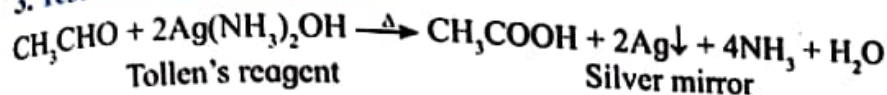


2. Test for Phenolic group



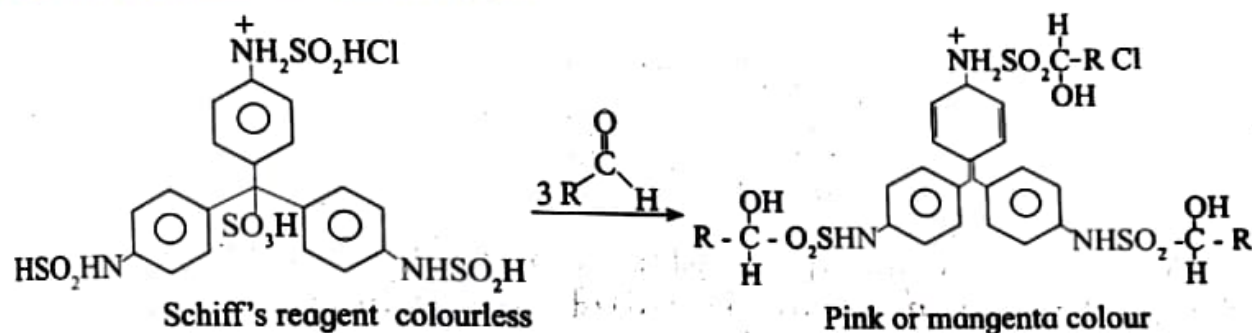
Imp note: 1. o-, m-, p- cresol and resorcinol gives violet or blue colouration.
2. β -naphthol gives green colouration. 3. α -naphthol gives pink colouration.

3. Test for aldehydic group



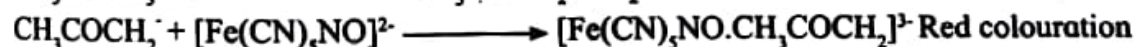
Tollen's reagent : Take 1 ml AgNO_3 + 2-3 ml $\text{NaOH} \longrightarrow$ forms brown precipitate. Add NH_3 dropwise till brown precipitate just dissolves.

Chemical equation for Schiff's reagent

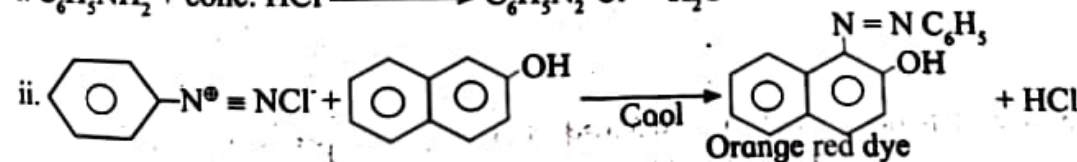
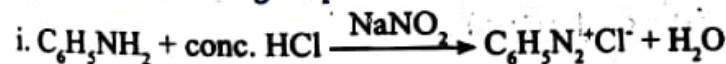


4. Test for Ketonic group

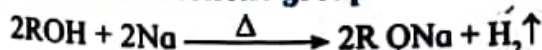
With sodium nitroprusside -



5. Test for Amino group



6. Test for alcoholic group



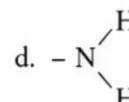
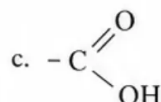
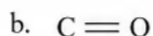
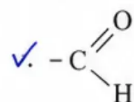
Organic substance to be given for functional group detection

Carboxylic acids: Acetic acid, oxalic acid, benzoic acid.	Phenols : β -naphthol, resorcinol, phenol
Aldehyde : Benzaldehyde, formaldehyde	Ketone : Acetone, ethylmethyl ketone.
Primary amines : p-toluidine, p-nitroaniline, aniline	Unsaturated hydrocarbons : Naphthalene, butadiene, cinnamic acid
Alcohols : ethyl alcohol.	

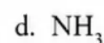
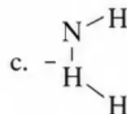
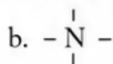
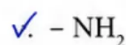
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Select [✓] the most appropriate answer from given alternatives of each sub question.

1. The functional group present in Methanal is



2. Identify the functional group of primary amine.



3. The distinguishing test between alcohol and phenol is -

a. bicarbonate test

✓. neutral FeCl_3 test

c. sodium nitroprusside test

d. tollen's reagent

4. Primary amines on treatment with excess of NaNO_2 , concentrated HCl and β -naphthol in alkaline medium gives.....

a. green colour

✓. orange colour

c. violet colour

d. red colour

5. Acetaldehyde when heated with ammonical silver nitrate solution gives

a. red ppt

✓. silver mirror

c. red colour

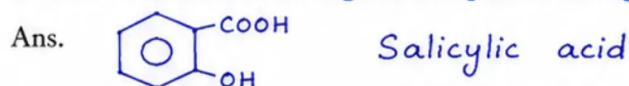
d. grey mirror

Short answer questions

1. Define: Functional group.

Ans. An atom or group of atom which determine chemical properties of molecule is called functional group.

2. Draw structure of an organic compound having two functional group in molecule.



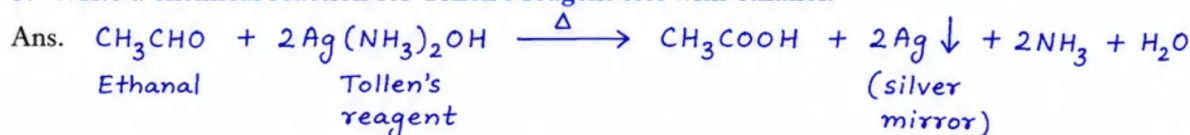
3. Give distinguishing test for carboxylic acid and phenol.

Ans. Sodium bicarbonate readily reacts with carboxylic acid but it does not react with phenol.

4. If alkaline KMnO_4 solution is added to an organic compound, it decolorises. Identify the functional group present in the compound?

Ans. If alkaline KMnO_4 solution is added to an organic compound, it decolorises. It indicates that such compound contain unsaturation, i.e. $\text{C}=\text{C}$ or $\text{C}\equiv\text{C}$.

5. Write a chemical reaction for Tollen's reagent test with ethanol.



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